

Year 7 · Chapter 2 · Angles: answers (teacher copy)

2.01 Naming angles (booklet p. 3)

Examples (model these live)

E1 Name the marked angle.

vertex in the middle · $\angle PQR$ or $\angle RQP$

E2 Name the vertex and the arms of $\angle XYZ$.

vertex: Y · arms: YX and YZ

E3 Name the three angles at O.

small: $\angle AOB$, $\angle BOC$ · whole: $\angle AOC$

Questions (31 on today's skill, 10 from earlier lessons)

- | | | | |
|--------------------------------|---|---|---|
| 1 U | 13 $\angle DMB$ or $\angle BMD$ | 23 $16^{1.06}$ | 33 $= 9 - 23 = -14^{1.08}$ |
| 2 $\angle YQC$ or $\angle CQY$ | 14 $\angle ECX$ | 24 $\angle HVW$ and $\angle WVG$ | 34 $21^\circ + 26^\circ = 47^\circ$ |
| 3 $\angle VPW$ | 15 vertex: H · arms: HC and HB | 25 $= 12 - 17 = -5^{1.04}$ | 35 6 angles · $\angle PJC$, $\angle CJY$, $\angle YJZ$ · $\angle PJY$, $\angle CJZ$, $\angle PJZ$ |
| 4 E | 16 small: $\angle RTQ$, $\angle QTB$ · whole: $\angle RTB$ | 26 vertex: N · arms: NS and NX | 36 6 angles · $\angle AOK$, $\angle KOW$, $\angle WOG$ · $\angle AOW$, $\angle KOG$, $\angle AOG$ |
| 5 $\angle BZA$ or $\angle AZB$ | 17 $-1^{1.02}$ | 27 $42^\circ + 52^\circ = 94^\circ$ | 37 No. They have different vertices (H and Q). |
| 6 $\angle LPG$ | 18 CP | 28 $13^{1.05}$ | 38 $6^{1.02}$ |
| 7 V | 19 $\angle UBL$ and $\angle LBF$ | 29 $46^\circ + 49^\circ = 95^\circ$ | 39 $60^\circ + 27^\circ = 87^\circ$ |
| 8 $\angle AZM$ or $\angle MZA$ | 20 $(-22)^2 = 484^{1.09}$ | 30 6 angles · $\angle COW$, $\angle WOG$, $\angle GOU$ · $\angle COG$, $\angle WOU$, $\angle COU$ | 40 bracket: $-168 = 2\ 184^{1.09}$ |
| 9 $\angle NKX$ | 21 small: $\angle MDS$, $\angle SDN$ · whole: $\angle MDN$ | 31 $5^{1.03}$ | 41 $\angle ZVR$ and $\angle RVE$ |
| 10 M | 22 RN | 32 No. They have different vertices (N and E). | |
| 11 $26^{1.01}$ | | | |
| 12 vertex: S · arms: SD and SU | | | |

2.02 Measuring and drawing angles (booklet p. 5)

Examples (model these live)

E1 Measure the angle.

0° on the bottom arm · $= 55^\circ$

E2 Draw an angle of 130° on the ray.

centre on the end, 0° on the ray · mark 130° , rule the arm

E3 How do you draw a reflex angle of 250° ?

$360^\circ - 250^\circ = 110^\circ$ · draw 110° , mark the outside

Questions (28 on today's skill, 9 from earlier lessons)

- | | | | |
|---|--|--|---|
| 1 155° | 13 estimate: about 110° · measure: 114° | 22 $360^\circ - 275^\circ = 85^\circ$ · draw 85° , mark the outside | 31 $360^\circ - 30^\circ = 330^\circ$ · 330° is reflex |
| 2 $360^\circ - 200^\circ = 160^\circ$ | 14 45° (check with a protractor) | 23 $360^\circ - 230^\circ = 130^\circ$ · draw 130° , mark the outside | 32 each number: $360^\circ \div 12 = 30^\circ$ · $2 \times 30^\circ = 60^\circ$ |
| 3 more | 15 $\angle UTA$ and $\angle ATZ^{2.01}$ | 24 $360^\circ - 70^\circ = 290^\circ$ · 290° is reflex | 33 10 apart, half is $5 \cdot 5$ right of -10 : $-5^{1.02}$ |
| 4 25° | 16 inside angle: $75^\circ \cdot 360^\circ - 75^\circ = 285^\circ$ | 25 $\angle MQK$ and $\angle KQA^{2.01}$ | 34 Read the wrong scale. The angle is 65° . |
| 5 $360^\circ - 225^\circ = 135^\circ$ | 17 135° (check with a protractor) | 26 each number: $360^\circ \div 12 = 30^\circ$ · $6 \times 30^\circ = 180^\circ$ | 35 $\angle MWB$ or $\angle BWM^{2.01}$ |
| 6 less | 18 inside angle: $165^\circ \cdot 360^\circ - 165^\circ = 195^\circ$ | 27 $4^{1.03}$ | 36 $360^\circ - 205^\circ = 155^\circ$ · draw 155° , mark the outside |
| 7 $360^\circ - 290^\circ = 70^\circ$ | 19 $\angle EFZ$ or $\angle ZFE^{2.01}$ | 28 Read the wrong scale. The angle is 25° . | 37 inside angle: $160^\circ \cdot 360^\circ - 160^\circ = 200^\circ$ |
| 8 more | 20 estimate: about 130° · measure: 131° | 29 $360^\circ - 220^\circ = 140^\circ$ · draw 140° , mark the outside | |
| 9 20° | 21 small: $\angle THP$, $\angle PHS$ · whole: $\angle THS^{2.01}$ | 30 $\angle XDB^{2.01}$ | |
| 10 $61 - 69 = -8$ dollars 1.10 | | | |
| 11 estimate: about 110° · measure: 109° | | | |
| 12 $360^\circ - 320^\circ = 40^\circ$ | | | |

2.03 Classifying angles (booklet p. 7)

Examples (model these live)

E1 Classify 135° .

between 90° and 180° · obtuse

E2 Classify 270° .

between 180° and 360° · reflex

E3 Classify the angle between the clock hands at 4:00.

$4 \times 30^\circ = 120^\circ$ · obtuse

Questions (36 on today's skill, 13 from earlier lessons)

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|--------------------------------------|---|---|--|
| 1 reflex | 10 between 180° and 360° | 19 inside angle: $120^\circ \cdot 360^\circ - 120^\circ = 240^\circ^{2.02}$ | 26 $\angle FVP^{2.01}$ |
| 2 acute | 11 acute | 20 $180^\circ + 85^\circ = 265^\circ$ · reflex | 27 $360^\circ - 205^\circ = 155^\circ$ · obtuse |
| 3 between 90° and 180° | 12 $-1^{1.04}$ | 21 $360^\circ - 210^\circ = 150^\circ$ · obtuse | 28 estimate: about 40° · measure: $44^\circ^{2.02}$ |
| 4 reflex | 13 $15^\circ + 45^\circ = 60^\circ$ · acute | 22 $1^{1.03}$ | 29 $35^\circ + 165^\circ = 200^\circ$ · reflex |
| 5 acute | 14 obtuse | 23 $15^\circ + 10^\circ = 25^\circ$ · acute | 30 Yes. Each is more than 90° , so the total is more than 180° . |
| 6 between 0° and 90° | 15 revolution | 24 $4 \times 30^\circ = 120^\circ$ · obtuse | |
| 7 straight | 16 $\angle ULV$ or $\angle VLU^{2.01}$ | 25 $90^\circ + 60^\circ = 150^\circ$ · obtuse | |
| 8 right | 17 $30^\circ + 75^\circ = 105^\circ$ · obtuse | | |
| 9 acute | 18 $2 \times 30^\circ = 60^\circ$ · acute | | |

- 31 total: $-20 \div -20 \div 4 \div = -5^\circ \text{C}$ ^{1.07}
- 32 $4 \times 30^\circ = 120^\circ \cdot$ obtuse
- 33 No. The obtuse angle alone is more than 90° .
- 34 $5 \times 45^\circ = 225^\circ \cdot$ reflex
- 35 160° (check with a protractor) ^{2.02}
- 36 $360^\circ \div 6 = 60^\circ \cdot$ acute
- 37 $= 36 - 6 \cdot = 30$ ^{1.08}
- 38 small angle: $90^\circ \cdot 360^\circ - 90^\circ = 270^\circ$
- 39 No. Each is less than 90° , so the total is less than 180° .
- 40 $\angle \text{VAP}$ or $\angle \text{PAV}$ ^{2.01}
- 41 $1.5 \times 90^\circ = 135^\circ \cdot$ obtuse
- 42 $360^\circ \div 4 = 90^\circ \cdot$ right
- 43 $((-) 22) x^2 = \cdot = 484$ ^{1.09}
- 44 small angle: $30^\circ \cdot 360^\circ - 30^\circ = 330^\circ$
- 45 $\angle \text{SBW}$ ^{2.01}
- 46 $360^\circ - 310^\circ = 50^\circ \cdot$ acute
- 47 $3 \times 60^\circ = 180^\circ \cdot$ straight
- 48 estimate: about $40^\circ \cdot$ measure: 41° ^{2.02}
- 49 $360^\circ \div 8 = 45^\circ \cdot$ acute

2.04 Complementary and supplementary angles (booklet p. 9)

Examples (model these live)

E1 Find the complement of 35° .
 $90^\circ - 35^\circ \cdot = 55^\circ$

E2 Find the supplement of 112° .
 $180^\circ - 112^\circ \cdot = 68^\circ$

E3 Find x.
 $3x + 60 = 180 \cdot 3x = 120 \cdot x = 40$

Questions (33 on today's skill, 11 from earlier lessons)

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|---|---|--|--|
| 1 24° | 14 $m + 73 = 90 \cdot m = 17$ | 27 inside angle: $155^\circ \cdot 360^\circ - 155^\circ = 205^\circ$ ^{2.02} | 36 complement: $65^\circ \cdot$ supplement: $180^\circ - 65^\circ = 115^\circ$ |
| 2 110° | 15 58° | 28 $y + 56 = 90 \cdot y = 34$ | 37 $3x + 72 = 180 \cdot 3x = 108 \cdot x = 36$ |
| 3 no (they add to 100°) | 16 $360^\circ - 325^\circ \cdot = 35^\circ$ ^{2.02} | 29 $3x + 30 = 180 \cdot 3x = 150 \cdot x = 50$ | 38 $x + x + 20 = 90 \cdot 2x = 70 \cdot x = 35$ |
| 4 $a = 85$ | 17 $y + 50 = 90 \cdot y = 40$ | 30 total: $0 \cdot = 0 \div 3 \cdot = 0^\circ \text{C}$ ^{1.07} | 39 $x = 3(180 - x) \cdot 4x = 540 \cdot x = 135^\circ$ |
| 5 27° | 18 $x + 79 = 180 \cdot x = 101$ | 31 $3x + 42 = 180 \cdot 3x = 138 \cdot x = 46$ | 40 $= 17 - 22 \cdot = -5$ ^{1.04} |
| 6 62° | 19 small: $\angle \text{GSX}, \angle \text{XSR} \cdot$ whole: $\angle \text{GSR}$ ^{2.01} | 32 $x + x + 10 = 90 \cdot 2x = 80 \cdot x = 40$ | 41 complement: $23^\circ \cdot$ supplement: $180^\circ - 23^\circ = 157^\circ$ |
| 7 no (they add to 100°) | 20 $b + 30 = 90 \cdot b = 60$ | 33 $360^\circ - 320^\circ = 40^\circ \cdot$ acute ^{2.03} | 42 obtuse ^{2.03} |
| 8 $a = 150$ | 21 Yes. $149 + 31 = 180$. | 34 $x = 2(90 - x) \cdot 3x = 180 \cdot x = 60^\circ$ | 43 $3x + 102 = 180 \cdot 3x = 78 \cdot x = 26$ |
| 9 71° | 22 3 ^{1.03} | 35 21 ^{1.01} | 44 Yes. $132 + 48 = 180$. |
| 10 no (they add to 100°) | 23 $x + 51 = 180 \cdot x = 129$ | | |
| 11 $a = 30$ | 24 $b + 40 = 90 \cdot b = 50$ | | |
| 12 $47 - 105 \cdot = -58$ dollars ^{1.10} | 25 between 180° and 360° ^{2.03} | | |
| 13 54° | 26 Yes. $157 + 23 = 180$. | | |

2.05 Angles at a point and vertically opposite angles (booklet p. 11)

Examples (model these live)

E1 Find a.
 $a + 125 = 180 \cdot a = 55$

E2 Find b.
 $b + 130 + 110 = 360 \cdot b = 120$

E3 Find c. Give a reason.
 $c = 70 \cdot$ vertically opposite angles

Questions (29 on today's skill, 11 from earlier lessons)

- | | | | |
|--|---|--|--|
| 1 $a = 150$ | 13 $a + 55 + 85 = 360 \cdot a = 220$ | 22 $6x = 360 \cdot x = 60$ | 32 $\angle \text{NSY}$ and $\angle \text{YSL}$ ^{2.01} |
| 2 $b = 75$ | 14 $c + 55 + 40 = 180 \cdot c = 85$ | 23 $c + 30 + 25 = 180 \cdot c = 125$ | 33 $2x + 15 = 53 \cdot 2x = 38 \cdot x = 19$ |
| 3 $c = 360 - 85 - 95 \cdot c = 180$ | 15 small: $\angle \text{MZX}, \angle \text{KZB} \cdot$ whole: $\angle \text{MZB}$ ^{2.01} | 24 yes ^{2.04} | 34 $4x + 100 = 180 \cdot 4x = 80 \cdot x = 20$ |
| 4 $a = 70$ | 16 $b = 50$ (vert. opp.) $\cdot a = 180 - 50 = 130 \cdot c = 130$ (vert. opp.) | 25 $9x = 360 \cdot x = 40$ | 35 48 ^{1.06} |
| 5 $b = 135$ | 17 $c + 25 + 35 = 180 \cdot c = 120$ | 26 $2x + 15 = 83 \cdot 2x = 68 \cdot x = 34$ | 36 No. Angles on a straight line add to 180° , so it is 140° . |
| 6 $c = 360 - 65 - 40 \cdot c = 255$ | 18 $b = 120$ (vert. opp.) $\cdot a = 180 - 120 = 60 \cdot c = 60$ (vert. opp.) | 27 $7 - (-1) \cdot = 7 + 1 \cdot = 8^\circ \text{C}$ ^{1.05} | 37 $\angle \text{GSK}$ or $\angle \text{KSG}$ ^{2.01} |
| 7 $b = 100$ | 19 -1 ⁶³² ^{1.09} | 28 $4x + 132 = 180 \cdot 4x = 48 \cdot x = 12$ | 38 $4x = 360 \cdot x = 90$ |
| 8 $c = 360 - 45 - 50 \cdot c = 265$ | 20 $a + 70 + 95 = 360 \cdot a = 195$ | 29 more ^{2.02} | 39 $b = 60$ (vert. opp.) $\cdot a = 180 - 60 = 120 \cdot c = 120$ (vert. opp.) |
| 9 $a = 80$ | 21 $\angle \text{BGN}$ ^{2.01} | 30 No. Angles on a straight line add to 180° , so it is 130° . | 40 right ^{2.03} |
| 10 6 ^{1.01} | | 31 $9x = 360 \cdot x = 40$ | |
| 11 $a + 115 + 100 = 360 \cdot a = 145$ | | | |
| 12 $b = 55$ | | | |

2.06 Parallel and perpendicular lines (booklet p. 13)

Examples (model these live)

E1 Write with a symbol: AB is perpendicular to CD.
 $\perp$ means perpendicular $\cdot AB \perp CD$

E2 Name the parallel sides of rectangle PQRS.
 opposite sides $\cdot PQ \parallel SR, PS \parallel QR$

E3 Construct a line perpendicular to AB through P, a point on AB.
 arcs on AB either side of P $\cdot$ equal arcs cross; rule from P

Questions (31 on today's skill, 10 from earlier lessons)

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|------------------------|-----------------------------|---------------------------|--|
| 1 $HX \perp WY$ | 6 SE is perpendicular to WC | 10 parallel | 14 $\parallel$: $AB \parallel DC, AD \parallel BC \cdot \perp$: every corner, e.g. $AB \perp BC$ |
| 2 parallel | 7 $CR \perp WD$ | 11 TL is parallel to XN | 15 perpendicular |
| 3 LC is parallel to TV | 8 neither | 12 $PU \perp GD$ | 16 CD is parallel to HK |
| 4 $HB \perp DZ$ | 9 $b = 95$ ^{2.05} | 13 -465 ^{1.09} | 17 142° ^{2.04} |
| 5 perpendicular | | | |

18 $\parallel$: PQ $\parallel$ SR, PS $\parallel$ QR $\cdot \perp$: none
 19 90°
 20 small: $\angle HSE, \angle ESG$ · whole: $\angle HSG$ ^{2.01}
 21 two lines that meet at 90° , marked with a right-angle mark
 22 90°
 23 $\parallel$: EF $\parallel$ HG $\cdot \perp$: none
 24 $= -8 - 6 = -14$ ^{1.08}

25 compasses at V: mark two points on PQ · from each point, equal arcs that cross · rule from V through the crossing
 26 $360^\circ - 230^\circ = 130^\circ$ ^{2.02}
 27 90°
 28 $\parallel$: WX $\parallel$ ZY, WZ $\parallel$ XY $\cdot \perp$: none
 29 a line at 90° to the ray through its end point (check with a set square)

30 rule a line through V that cuts PQ · copy the angle it makes at V · rule the new arm: it is $\parallel$ to PQ
 31 total: $0 \cdot = 0 \div 4 \cdot = 0^\circ C$ ^{1.07}
 32 $AB \perp EF$
 33 between 180° and 360° ^{2.03}
 34 compasses more than half of PQ · arcs from P and from Q, above and below · rule through the two crossings

35 $\perp$ to WX: WZ and XY $\cdot \parallel$ to WX: ZY
 36 $AB \parallel EF$
 37 $\perp$ to AB: AD and BC $\cdot \parallel$ to AB: DC
 38 0 ^{1.05}
 39 $AB \parallel EF$
 40 less ^{2.02}
 41 $\perp$ to PQ: PS and QR $\cdot \parallel$ to PQ: SR

2.07 Corresponding angles (booklet p. 15)

Examples (model these live)

E1 The lines are parallel. Find a.
 corresponding angles are equal $\cdot a = 70$

E2 Find b, the angle next to a on the line.
 $b + 70 = 180$ (straight line) $\cdot b = 110$

E3 Corresponding angles are $(2x + 10)^\circ$ and 70° . Find x.
 $2x + 10 = 70 \cdot 2x = 60 \cdot x = 30$

Questions (29 on today's skill, 9 from earlier lessons)

1 $a = 135$
 2 140°
 3 q
 4 $a = 35$
 5 24°
 6 48°
 7 p
 8 $a = 135$
 9 -1 ^{1.31} ^{1.09}
 10 $a = 140$ (corresponding) · $b = 180 - 140 = 40$
 11 44°
 12 $a = 40$ (corresponding) · $b = 180 - 40 = 140$

13 $a = 150$ · corresponding angles, $\parallel$ lines
 14 small: $\angle UYF, \angle FYC$ · whole: $\angle UYC$ ^{2.01}
 15 $x + 5 = 25 \cdot x = 20$
 16 $a = 145$ (corresponding) · $b = 180 - 145 = 35$
 17 $a = 135$ · corresponding angles, $\parallel$ lines
 18 $x + 5 = 10 \cdot x = 5$
 19 $= -24 - 10 = -34$ ^{1.08}
 20 $a = 90$ (corresponding) · $b = 180 - 90 = 90$
 21 $360^\circ - 350^\circ = 10^\circ$ ^{2.02}
 22 $4x + 5 = 49 \cdot 4x = 44 \cdot x = 11$

23 $a = 60$ · corresponding angles, $\parallel$ lines
 24 $4x + 5 = 73 \cdot 4x = 68 \cdot x = 17$
 25 $y = 180 - 55 = 125$ (straight) · $x = 125$ (corresponding)
 26 12 apart, half is 6 · 6 right of -9: -3 ^{1.02}
 27 No. Corresponding angles on parallel lines are equal.
 28 yes ^{2.04}
 29 $2x + 25 = 61 \cdot 2x = 36 \cdot x = 18$

30 $y = 180 - 140 = 40$ (straight) · $x = 40$ (corresponding)
 31 $y = 180 - 85 = 95$ (straight) · $x = 95$ (corresponding)
 32 $3x + 5 = 74 \cdot 3x = 69 \cdot x = 23$
 33 $75 - 86 = -11$ dollars ^{1.10}
 34 $x + 5 = 19 \cdot x = 14$
 35 $\angle VXH$ ^{2.01}
 36 $y = 180 - 125 = 55$ (straight) · $x = 55$ (corresponding)
 37 $4x + 15 = 39 \cdot 4x = 24 \cdot x = 6$
 38 $ZK \parallel HX$ ^{2.06}

2.08 Alternate angles (booklet p. 17)

Examples (model these live)

E1 The lines are parallel. Find a.
 alternate angles are equal $\cdot a = 50$

E2 a is alternate to 115° . Find a, then b next to it on the line.
 $a = 115$ (alternate) $\cdot b = 180 - 115 = 65$

E3 Alternate angles are $(3x - 15)^\circ$ and 60° . Find x.
 $3x - 15 = 60 \cdot 3x = 75 \cdot x = 25$

Questions (29 on today's skill, 8 from earlier lessons)

1 $a = 135$
 2 106°
 3 between them
 4 $a = 125$
 5 49°
 6 Z (or N)
 7 32°
 8 opposite sides
 9 $a = 95$
 10 $70 - 79 = -9$ dollars ^{1.10}
 11 $a = 155$ (alternate) · $b = 180 - 155 = 25$
 12 88°

13 $a = 125$ (alternate) · $b = 180 - 125 = 55$
 14 $a = 120$ · alternate angles, $\parallel$ lines
 15 small: $\angle EVY, \angle YVW$ · whole: $\angle EVW$ ^{2.01}
 16 $x - 15 = 12 \cdot x = 27$
 17 $a = 30$ (alternate) · $b = 180 - 30 = 150$
 18 $a = 50$ · alternate angles, $\parallel$ lines
 19 $x - 25 = 0 \cdot x = 25$
 20 30 ^{1.01}

21 $a = 140$ (alternate) · $b = 180 - 140 = 40$
 22 $360^\circ - 275^\circ = 85^\circ$ ^{2.02}
 23 $3x - 15 = 30 \cdot 3x = 45 \cdot x = 15$
 24 $a = 130$ · alternate angles, $\parallel$ lines
 25 $2x - 25 = 29 \cdot 2x = 54 \cdot x = 27$
 26 $a = 35$ · corresponding
 27 $2 - (-3) = 2 + 3 = 5^\circ C$ ^{1.05}
 28 $p = 135$ (alternate) · $q = 180 - 135 = 45$ (straight)

29 $c = 360 - 65 - 40 \cdot c = 255$ ^{2.05}
 30 $2x - 5 = 35 \cdot 2x = 40 \cdot x = 20$
 31 $a = 30$ · alternate
 32 $p = 85$ (alternate) · $q = 180 - 85 = 95$ (straight)
 33 -5 ^{1.04}
 34 $4x - 5 = 107 \cdot 4x = 112 \cdot x = 28$
 35 38° ^{2.04}
 36 $x - 30 = 7 \cdot x = 37$
 37 $a = 45$ · alternate

2.09 Co-interior angles (booklet p. 19)

Examples (model these live)

E1 The lines are parallel. Find a.
 $a + 110 = 180$ (co-interior) $\cdot a = 70$

E2 a is co-interior to 65° . Find a.
 $a + 65 = 180 \cdot a = 115$

E3 Co-interior angles are y° and $2y^\circ$. Find y.
 $y + 2y = 180 \cdot 3y = 180 \cdot y = 60$

Questions (30 on today's skill, 11 from earlier lessons)

- | | | | |
|---------------------------------------|---|---|---|
| 1 $a = 55$ | 14 $a = 180 - 125 = 55 \cdot$ co-interior angles, $\parallel$ lines | 24 $a = 180 - 100 = 80 \cdot$ co-interior angles, $\parallel$ lines | 34 $2x + 20 + 126 = 180 \cdot 2x = 34 \cdot x = 17$ |
| 2 $180^\circ - 83^\circ = 97^\circ$ | 15 small: $\angle LDG, \angle GDA \cdot$ whole: $\angle LDA$ ^{2.01} | 25 opposite sides ^{2.08} | 35 $a = 50 \cdot$ co-interior |
| 3 add to 180° | 16 Yes. $102 + 78 = 180$. | 26 $y + 3y = 180 \cdot 4y = 180$, so $y = 45 \cdot$ angles: 45° and 135° | 36 25 ^{1.06} |
| 4 $a = 75$ | 17 $a + 47 = 180 \cdot a = 133$ | 27 $2x + 20 + 118 = 180 \cdot 2x = 42 \cdot x = 21$ | 37 $y + 8y = 180 \cdot 9y = 180$, so $y = 20 \cdot$ angles: 20° and 160° |
| 5 $180^\circ - 124^\circ = 56^\circ$ | 18 $a = 180 - 55 = 125 \cdot$ co-interior angles, $\parallel$ lines | 28 $= 2 - (-1) = 3$ ^{1.08} | 38 44° ^{2.04} |
| 6 yes | 19 No. $32 + 138 = 170$, not 180. | 29 $a = 125 \cdot$ corresponding | 39 No. $45 + 125 = 170$, not 180. |
| 7 $180^\circ - 148^\circ = 32^\circ$ | 20 -1 ^{1.09} | 30 SZ is parallel to NR ^{2.06} | 40 $2x + 10 + 148 = 180 \cdot 2x = 22 \cdot x = 11$ |
| 8 C (or U) | 21 $a + 39 = 180 \cdot a = 141$ | 31 $y + 4y = 180 \cdot 5y = 180$, so $y = 36 \cdot$ angles: 36° and 144° | 41 $a = 105$ ^{2.07} |
| 9 $a = 110$ | 22 $360^\circ - 230^\circ = 130^\circ$ ^{2.02} | 32 $2x + 30 + 106 = 180 \cdot 2x = 44 \cdot x = 22$ | |
| 10 -7 ^{1.02} | 23 $y + 5y = 180 \cdot 6y = 180$, so $y = 30 \cdot$ angles: 30° and 150° | 33 $1 \times 30^\circ = 30^\circ \cdot$ acute ^{2.03} | |
| 11 $a + 128 = 180 \cdot a = 52$ | | | |
| 12 $180^\circ - 34^\circ = 146^\circ$ | | | |
| 13 $a + 158 = 180 \cdot a = 22$ | | | |

2.10 Finding unknown angles on parallel lines (booklet p. 21)

Examples (model these live)

- | | | |
|---|--|---|
| E1 Which rule? Find a.
Z shape: alternate $\cdot a = 65$ | E2 Find b. Give a reason.
C shape: co-interior $\cdot b = 180 - 65 = 115$ | E3 Find y, then x. Give reasons.
$y = 180 - 50 = 130$ (straight) $\cdot x = 130$ (corresponding) |
|---|--|---|

Questions (26 on today's skill, 8 from earlier lessons)

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|---|--|--|---|
| 1 corresponding | 13 $a = 60 \cdot$ corresponding angles are equal | 20 $a = 80 \cdot$ alternate angles are equal | 27 q ^{2.07} |
| 2 $a = 100$ | 14 $y = 180 - 140 = 40$ (straight) $\cdot x = 40$ (corresponding) | 21 $360^\circ - 240^\circ = 120^\circ$ ^{2.02} | 28 $3x + 15 = 60 \cdot 3x = 45 \cdot x = 15$ |
| 3 are equal | 15 small: $\angle KZS, \angle SZX \cdot$ whole: $\angle KZX$ ^{2.01} | 22 $p = 85$ (corresponding, then vert. opp.) $\cdot q = 180 - 85 = 95$ (straight line) | 29 No. Co-interior angles add to 180° , so $a = 90$. |
| 4 alternate | 16 $m + 80 = 180 \cdot m = 100$ | 23 $p = 130$ (corresponding, then vert. opp.) $\cdot q = 180 - 130 = 50$ (straight line) | 30 -6 ^{1.03} |
| 5 $a = 150$ | 17 $y = 180 - 65 = 115$ (straight) $\cdot x = 115$ (corresponding) | 24 $3x + 15 = 75 \cdot 3x = 60 \cdot x = 20$ | 31 $p = 40$ (corresponding, then vert. opp.) $\cdot q = 180 - 40 = 140$ (straight line) |
| 6 are equal | 18 $m = 53 \cdot m = 53$ | 25 total: $-10 \cdot = -10 \div 5 \cdot = -2^\circ\text{C}$ ^{1.07} | 32 131° ^{2.04} |
| 7 co-interior | 19 $44 - 60 \cdot = -16$ dollars ^{1.10} | 26 No. Co-interior angles add to 180° , so $a = 100$. | 33 $m + 159 = 180 \cdot m = 21$ |
| 8 $a = 95$ | | | 34 $3x + 15 = 111 \cdot 3x = 96 \cdot x = 32$ |
| 9 add to 180° | | | |
| 10 corresponding | | | |
| 11 12 ^{1.05} | | | |
| 12 $a = 180 - 100 = 80 \cdot$ co-interior angles add to 180° | | | |

2.11 Proving lines are parallel (booklet p. 23)

Examples (model these live)

- | | | |
|--|--|--|
| E1 Alternate angles are 64° and 64° .
Is $AB \parallel CD$?
$64 = 64 \cdot$ yes: alternate angles are equal | E2 Co-interior angles are 105° and 85° . Is $AB \parallel CD$?
$105 + 85 = 190 \cdot$ no: they do not add to 180° | E3 Co-interior angles are $2x^\circ$ and 120° . Find x so that $AB \parallel CD$.
$2x + 120 = 180 \cdot 2x = 60 \cdot x = 30$ |
|--|--|--|

Questions (34 on today's skill, 13 from earlier lessons)

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|--|--|---|---|
| 1 yes | 16 $125 + 55 = 180 \cdot$ yes: co-interior angles add to 180° | 26 $3x + 10 = 88 \cdot 3x = 78 \cdot x = 26$ | 37 $3x + 10 = 85 \cdot 3x = 75 \cdot x = 25$ |
| 2 they are equal | 17 $180^\circ - 36^\circ = 144^\circ$ ^{2.09} | 27 $360^\circ - 210^\circ = 150^\circ$ ^{2.02} | 38 No. $80 + 105 = 185$, not 180. |
| 3 yes | 18 $30 = 30 \cdot$ yes: corresponding angles are equal | 28 $2x + 122 = 180 \cdot 2x = 58 \cdot x = 29$ | 39 $5 \times 30^\circ = 150^\circ \cdot$ obtuse ^{2.03} |
| 4 they are equal | 19 $2x + 134 = 180 \cdot 2x = 46 \cdot x = 23$ | 29 $80 \neq 85 \cdot$ no: they are not equal | 40 No. $60 + 125 = 185$, not 180. |
| 5 yes | 20 small: $\angle TWY, \angle YWV \cdot$ whole: $\angle TWV$ ^{2.01} | 30 are equal ^{2.10} | 41 No. Co-interior angles must add to 180° ; $115 + 115 = 230$. |
| 6 they add to 180° | 21 $30 = 30 \cdot$ yes: alternate angles are equal | 31 $3x + 10 = 34 \cdot 3x = 24 \cdot x = 8$ | 42 2 ^{1.02} |
| 7 yes | 22 $2x + 130 = 180 \cdot 2x = 50 \cdot x = 25$ | 32 $3x + 10 = 58 \cdot 3x = 48 \cdot x = 16$ | 43 $3x + 10 = 40 \cdot 3x = 30 \cdot x = 10$ |
| 8 yes | 23 $2x + 148 = 180 \cdot 2x = 32 \cdot x = 16$ | 33 No. $70 + 115 = 185$, not 180. | 44 163° ^{2.04} |
| 9 $a = 90$ ^{2.10} | 24 $75 = 75 \cdot$ yes: corresponding angles are equal | 34 $= -12 - 5 \cdot = -17$ ^{1.08} | 45 $2x + 146 = 180 \cdot 2x = 34 \cdot x = 17$ |
| 10 yes | 25 -28 ^{1.01} | 35 No. Co-interior angles must add to 180° ; $125 + 125 = 250$. | 46 No. $55 + 130 = 185$, not 180. |
| 11 $60 = 60 \cdot$ yes: corresponding angles are equal | | 36 Z (or N) ^{2.08} | 47 $a = 55$ ^{2.09} |
| 12 yes | | | |
| 13 -5 ^{1.03} | | | |
| 14 $100 = 100 \cdot$ yes: alternate angles are equal | | | |
| 15 yes | | | |